12.Hardware troubleshooting & capstone introduction :

1.Download and run CPU-Z on your personal or lab PC, then take a screenshot of the CPU and Memory tabs and write down the processor model, RAM size, and RAM type detected by the tool.

ANS:

1. Download **CPU-Z** from the official website: CPU-Z official download
2. Install and open **CPU-Z**.
3. Open the **CPU** tab → take a screenshot.
4. Open the **Memory** tab → take a screenshot.
5. From the screenshots, note:
    - **Processor Model** → CPU tab → *Name*
    - **RAM Size** → Memory tab → *Size*
    - **RAM Type** → Memory tab → *Type* (e.g., DDR4/DDR5)

Write in your practical

Item	**Detected Information**
Processor Model	_____
RAM Size	_____
RAM Type	_____

**2.Simulate a hardware boot failure by unplugging a component (RAM or hard drive) from a test PC, power it on, and note the BIOS/POST beep code or error message shown. Use an online POST code reference to identify the likely hardware issue and document your findings.

Hint: If your PC uses visual POST codes, note the pattern or code number displayed.**

ANS:

What you need to do

1. Use a **test PC in the lab**.
2. With the PC powered off, the instructor can disconnect **RAM** (or another permitted component).
3. Power on the PC.
4. Observe what happens:
    - Beep pattern from the motherboard speaker, **or**
    - LED/POST code, **or**
    - Error message on the screen.
5. Take a photo/screenshot of the error if permitted.
6. Look up the motherboard manufacturer's POST/beep-code documentation and identify the likely problem.

Practical write-up format

Experiment: Simulate Hardware Boot Failure

Item	**Observation**
Component disconnected	RAM
POST result	PC failed to boot
Beep/POST code	_____
Error message	_____
Identified issue	RAM not detected / memory-related POST failure
Reference used	Motherboard manufacturer's POST/beep-code documentation

3. Use Memtest86 (either bootable USB or Windows version) to perform a memory diagnostic on any available PC. Record any errors found and explain what kind of real-world problems (e.g., random crashes, blue screens) these errors could cause.

ANS:

Steps

1. Download **MemTest86** from the official site: MemTest86 official website
2. Create a **bootable USB** using the MemTest86 image/installer.
3. Restart the PC and boot from the USB.
4. MemTest86 will start the memory test automatically.
5. Let the test complete, preferably for at least **one full pass**.
6. Record the **Pass** and **Errors** values.
7. Take a screenshot/photo of the result screen.

Practical write-up

Item	**Result**
Tool Used	MemTest86
Test Performed	RAM / Memory Diagnostic
Number of Passes	_____
Errors Found	_____
RAM Status	_____

If your test shows 0 errors

Result:
MemTest86 completed the memory diagnostic with **0 errors**. No memory errors were detected during the test.

If errors are found

Memory errors can cause problems such as:

- Random application crashes
- System freezes
- Blue Screen of Death (BSOD)
- Unexpected restarts
- Corrupted files or data
- Programs behaving unpredictably

4. Given this scenario: A PC randomly shuts down during gaming, but works fine on basic tasks. Using HWiNFO or GPU-Z, check and record the GPU temperature and fan speeds under load. Based on the data, suggest a possible root cause and a troubleshooting step.

Use HWiNFO or GPU-Z while running a game or GPU stress test. Record the GPU temperature and fan speed when the shutdown happens or just before it.

Parameter	Observation
GPU Temperature under load	_____ °C
GPU Fan Speed	_____ RPM / _____ %
Problem observed	PC randomly shuts down during gaming
Possible Root Cause	GPU overheating
Troubleshooting Step	Clean dust from the GPU/cooling system and check that the GPU fans are working properly

Explanation

If the GPU temperature becomes unusually high under gaming load while basic tasks work normally, GPU overheating is a possible cause. The system may shut down as a protective measure.

Troubleshooting: Clean the GPU heatsink/fans, make sure airflow is unobstructed, and verify that the GPU fans increase their speed under load.

**5.Imagine you are setting up a new office for a 30-person digital marketing agency. Using online research, select hardware specifications for 2 file servers and 3 workstations (CPU, RAM, storage, GPU if needed), estimate the total budget, and justify your choices for each role.

Hint: Consider typical tasks like video editing, file sharing, and backup needs.**

ANS:

For a 30-person digital marketing agency, I would separate the hardware into shared file/backup servers and high-performance creative workstations. Adobe currently recommends 32 GB+ RAM for 4K video work, 8 GB GPU memory, fast SSD storage, and 10-Gigabit networking for 4K shared workflows.

1. File Servers — 2 units

Component	Specification	Approx. cost/unit
CPU	AMD Ryzen R1600-class NAS CPU	Included
RAM	4 GB ECC DDR4, expandable	Included
Storage	4 × 8 TB NAS HDD, RAID-5/SHR	₹80,000
Network	1GbE initially; 10GbE upgrade	₹10,000
Purpose	File sharing + backup	—
Estimated/unit		₹90,000
2 servers		₹1,80,000

A Synology DS923+ is an example of this type of NAS: it has a 4-core/2-thread-class AMD R1600 platform, ECC memory support, four SATA drive bays and a PCIe slot for networking upgrades.





Why 2 servers?

- Server 1 → daily shared company files/projects.
- Server 2 → backup/secondary copy.
- If one server fails, the company still has another copy of important data.
- RAID protects against an individual drive failure, but RAID is not a substitute for backup.

2. Video/Creative Workstations — 3 units

Component	Specification	Approx. cost/unit
CPU	Intel Core i7 / AMD Ryzen 7, 8+ cores	₹35,000
RAM	32 GB DDR5	₹10,000
GPU	NVIDIA RTX 4060 8 GB	₹30,000
Storage	1 TB NVMe SSD + 2 TB SSD	₹15,000
Motherboard/PSU/Case	Suitable workstation components	₹30,000

Component	Specification	Approx. cost/unit
Estimated/unit		₹1,20,000
3 workstations		₹3,60,000

The 32 GB RAM and 8 GB GPU target is deliberate: Adobe recommends 32 GB or more for 4K+ editing and 8 GB GPU memory for current Premiere workflows.

Intel Core i5 + RTX 4060 workstation
₹1,19,899•Powerful i5 workstation suitable for multitasking and gaming.

Core i9 + RTX 4060 workstation
₹1,54,990•High-performance i9 workstation with advanced graphics for demanding tasks.
Give feedback

3. Total Estimated Budget

Hardware	Quantity	Estimated total
File servers	2	₹1,80,000
Creative workstations	3	₹3,60,000
Total	5 systems	₹5,40,000

Justification

File servers: High-capacity NAS storage with redundancy is appropriate because a 30-person agency will generate large amounts of marketing assets, videos, images and project files. Two servers also allow one to act as the primary file server and the other as a backup target.
Workstations: The three higher-performance PCs are intended for employees doing video editing, graphics, motion content and other demanding creative work. The RTX-class GPU provides hardware acceleration for supported video-editing operations, while 32 GB RAM gives enough headroom for 4K editing and multitasking.